

EXPERIMENT 17: Prolog Program to Sum Integers from 1 to N

Aim

To write and implement a Prolog program to find the sum of integers from 1 to N using recursion.

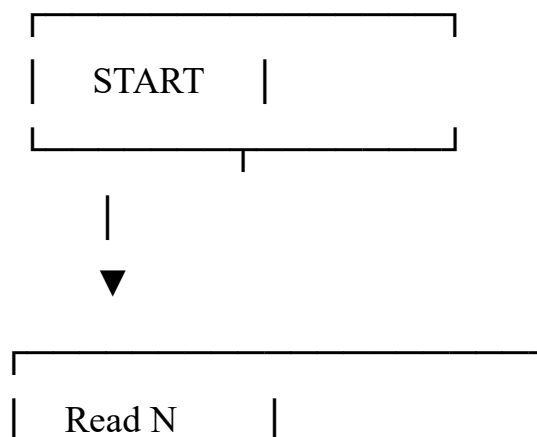
Objective

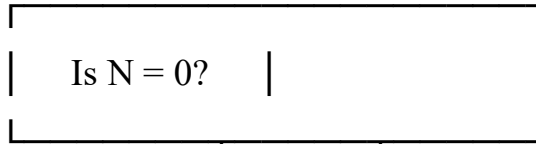
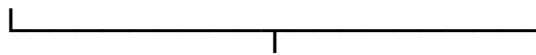
- To understand recursion in Prolog.
- To calculate the sum of integers from 1 to N.
- To understand facts, rules, and arithmetic operations in Prolog.
- To execute a simple mathematical problem using Prolog.

Algorithm

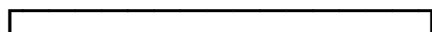
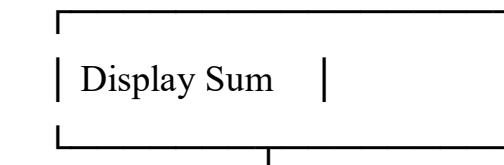
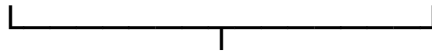
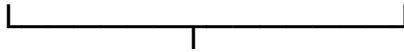
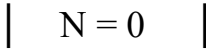
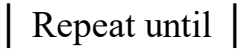
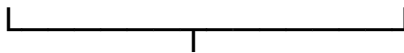
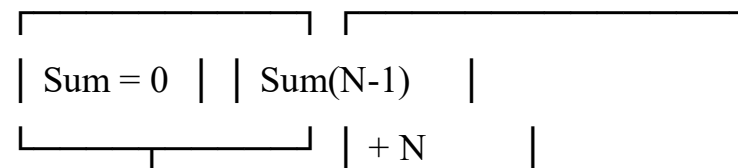
1. Start the program.
2. Read the value of N.
3. If $N = 0$, return 0.
4. Otherwise, calculate the sum of integers from 1 to $N-1$.
5. Add N to the calculated sum.
6. Repeat the process recursively until $N = 0$.
7. Display the final sum.
8. Stop.

Flowchart





Yes No



```
|   END   |  
└─────────┘
```

Code

% Prolog Program to Sum Integers from 1 to N

% Base case

sum(0, 0).

% Recursive case

sum(N, S) :-

N > 0,

N1 is N - 1,

sum(N1, S1),

S is N + S1.

Query

?- sum(10, S).

Output

S = 55.

Another example:

?- sum(5, S).

Output:

```
main.py
1  # Python Program to Sum Integers from 1 to N
2
3  # Read N from the user
4  N = int(input("Enter the value of N: "))
5
6  # Initialize sum
7  total = 0
8
9  # Calculate sum using a loop
10 for i in range(1, N + 1):
11     total = total + i
12
13 # Display the result
14 print("Sum of integers from 1 to", N, "is:", total)
15
```

input

Enter the value of N: 5
Sum of integers from 1 to 5 is: 15

Result

The Prolog program was successfully implemented to calculate the sum of integers from 1 to N using recursion.

For example, when $N = 10$:

$$1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 = 55$$

Therefore, the output is 55.

Conclusion

The Prolog program successfully calculates the sum of integers from 1 to N. This program demonstrates the use of recursion, arithmetic expressions, rules, and base cases in Prolog.